

Education

University of Maryland

Mathematics (B.S.) and Economics (B.S.), GPA: 3.65

January 2021 – December 2024

College Park, Maryland

Work Experience

Data Scientist

January 2025 – Present

Bethesda, MD

- Built high-throughput **C#** hex decoder and **Parquet** exporter for classified maintenance sensor data, converting hex files into analytics-ready columnar output at **2,656** files per minute, about **1 GB/s**, using multithreaded processing. Resulting **Parquet** tables were **85%** smaller than the raw hex data.
- Designed a scalable **Docker Compose/NixOS ETL** data pipeline with **C#** dependency injection, a central distributor, worker containers, and **NATS** messaging/observability; processed over **4 TB** of production records with minimal manual intervention.
- Built a **Rust** datetime extrapolation library, integrated via **FFI** into an **ETL** pipeline, that corrected on average **97%** of temporal data inconsistencies, utilizing **Polars** and **chrono**.
- Built open-source **Nix** library nix-nuget-feed, repackaging Nix-derived **NuGet** packages into local feeds for reproducible **.NET** development environments.
- Conducted correlation analysis on engineering configurations and error alerts using **sklearn** decision trees. Presented findings to government customers. Identified an **85%** increase in error occurrence associated with a specific configuration.

Statistician Internship

USDA, National Agricultural Statistical Service

May 2023 – December 2024

Harrisburg, PA

- Conducted accuracy testing of the JIMMY tool designed to automate the correction of survey data errors using **Python** and **Pandas**, comparing outputs against traditional NASS data processing.
- Developed a user interface with **Tkinter** to highlight discrepancies in survey variables, including over-estimations and under-estimations by JIMMY, and implemented **Bland-Altman plots** to enhance stakeholder understanding of variable comparisons.

Self-Directed Projects

REAPER Home Manager Flake

GitHub:9Prestidigitator/reaper-flake

October 2025 – Present

- Built a **Nix Home Manager** module for **REAPER** that maps plaintext `.ini` state into declarative options for reproducible digital audio workstation configuration across **NixOS** hosts.
- Wrote **Python** activation script to mask and merge **REAPER** bitfield-backed settings, preserving user-managed bits while applying Nix-managed configuration safely across rebuilds.
- Integrated the module into a flake-based workstation configuration with pinned dependencies, overlays, development shells, and audio production-specific host options.

Linux Audio Packages

GitHub:9Prestidigitator/maxpkg

April 2026 – Present

- Maintained a **Nix Flake** exposing over **39** Linux audio package derivations, an overlay, and a **NixOS** module for tools missing from nixpkgs or requiring newer upstream versions.
- Packaged native audio applications and plugins across CLAP, LV2, VST2, and VST3 formats, including patched **C++/CMake** builds with ONNX Runtime, TensorFlow, and JACK/PipeWire integration.

JUCE EQ Audio Plugin

GitHub:9Prestidigitator/juce_eq

July 2025

- Created audio equalizer VST plugin in **C++** using **JUCE** framework for UI and digital signal processing.
- Implemented low cut, high cut (Butterworth filters), and parametric peak filters with adjustable frequency, gain, and slope controls.

Technical Skills

Languages: C#, Nix, Rust, C++, Python, Java, Bash, SQL, R, and Lua.

Tooling: Neovim, Git, tmux, Docker, Nix Flakes, just, $\text{\LaTeX}$, Codex, hermes-agent, uv, Obsidian and MATLAB.

Frameworks/Libraries: .NET, Parquet, Polars, ndarray, Rayon, chrono, clap, NATS, stateless, Serilog, flake-parts, Home Manager and JUCE.

Operating Systems: NixOS, macOS, Arch Linux, Debian, and Windows 11.